

NPLify · P0 Technical Baseline

Deliverable 3 — Calculation Specification · Draft v1.0

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Status: Built against ERD & Data Model Draft v4.0 and the Project Understanding v1.0 with NPL's review comments. This document states every calculation the engine performs as a precise formula, in the order it is performed, with the rounding rule that applies at each step, and gives reference vectors computed from those formulas. It is the acceptance yardstick: the engine passes when it reproduces every vector to the cent.

What it does not yet contain: NPL's reference workbooks per project archetype (open question 11) and the reconciliation of the FRS worked figures (open question 9). When they arrive, their cases are added to Section 9 as further vectors and the formulas are confirmed or corrected against them.

1 · Conventions

- **Rates** are written as *OUT units per 1 IN unit* (EUR per USDT for a USDT → EUR deal). A pair configured the other way round (`quote_direction = in_per_out`) is inverted once, at full precision, before any formula below is applied: $R = 1 / R_{\text{quoted}}$.
- **Percentages** are decimals in formulas (0.40 % is 0.004).
- **Arithmetic** is exact decimal arithmetic, never binary floating point. Intermediate results are not rounded. Rounding happens only where this document says so.
- **Symbols:** A_{in} amount collected (IN currency); R_{m} market reference rate; R_{p} partner rate; R_{b} basis rate; s sender share; r receiver share; $f = s + r$ total client-side fee; m partner markup where the partner states $\text{market} + m$.

2 · Fee basis

The fee structure for a partner × pair names a **basis**: the rate NPL's price is built on.

```
R_b = R_m    if rate_basis = market
R_b = R_p    if rate_basis = partner
```

With *partner* basis NPL's margin is exactly the client-side fee, whatever the partner's spread. With *market* basis the partner's spread against market eats into the margin, which is why the minimum-margin check (Section 7) exists and why variable fee mode (Section 4) adjusts the fee to the spread.

3 · Sender and receiver split

The total fee f is split into a sender share s , embedded in the rate the sender sees, and a receiver share r , deducted from what the receiver is entitled to. The two are applied one after the other, so the client net is $A_{\text{in}} \times R_{\text{b}} \times (1 - s) \times (1 - r)$, slightly more than $A_{\text{in}} \times R_{\text{b}} \times (1 - f)$ by the cross term $s \times r$.

sender_rate	R_{s}	$= R_{\text{b}} \times (1 - s)$	stored to the pair's
rate_precision			
gross_out		$= \text{round2}(A_{\text{in}} \times R_{\text{s}})$	what the sender's
price promises			

```

fee_receiver_part      = round2( gross_out × r )
client_net             = gross_out - fee_receiver_part      the receiver group's
entitlement
fee_sender_part        = round2( A_in × R_b ) - gross_out    the sender share, in
the OUT currency

```

round2 rounds half up to the OUT currency's minor unit (2 decimals for fiat; 6 for USDT; 8 for BTC). Where the deal has a per-sender FEE_OVERRIDE, its pct replaces f and is split in the structure's s : r proportion. Where an introducer is paid a fixed percentage inside the sender-facing markup, that percentage is added to s before the sender rate is computed (Section 8).

4 · Fixed and variable fee modes

Fixed: f is the structure's pct. **Variable:** f is set per deal to protect the margin floor against the partner's current spread, within the structure's floor and cap.

```

partner_spread σ = 1 - R_p / R_m          what the partner keeps relative
to market
required      = min_margin_pct + σ      if rate_basis = market
              = min_margin_pct          if rate_basis = partner
f              = clamp( required, floor_pct, cap_pct )
below_margin   = ( f < required )        → the quote proceeds only with Finance approval
split          s = f × sender_share_pct / pct ; r = f - s      (same proportion as the
structure)

```

Case	R_m	R_p	Basis	Spread σ	Required	f (floor 0.40 %, cap 1.50 %, min margin 0.50 %)	Below margin?
Normal day	0.9	0.895	market	0.556 %	1.056 %	1.056 %	no
Partner spread widens to 2.22 %	0.9	0.88	market	2.222 %	2.722 %	1.500 %	yes, Finance approval
Partner basis: spread does not matter	0.9	0.895	partner	0.556 %	0.500 %	0.500 %	no

5 · What the partner delivers, and NPL's earnings

```

disclosed-rate partner (margin inside its rate):
  gross_converted = round2( A_in × R_p )
  partner_cost    = 0 in the books; inferred_cost = round2( A_in × (R_m - R_p) ) kept as
information
market + m partner (fee stated separately):
  gross_converted = round2( A_in × R_m )
  partner_cost    = round2( gross_converted × m )
in both cases:
  actual_out      = gross_converted - partner_cost          what the partner books for
the client
  gross_margin    = gross_converted - client_net            = partner_cost + NPL earnings
earnings          = actual_out - client_net                NPL's margin, recognised at
conversion
margin_pct        = earnings / actual_out

```

When the partner calculates (calc_mode = partner_calculates), actual_out is whatever the partner booked; expected_out is the formula above; variance = expected_out - actual_out and an exception opens if |variance|

/ expected_out exceeds the project's rate-variation tolerance. The client net never changes: the sender's price is fixed, so a variance is NPL's cost or gain.

6 · Target-out (quote-first) and collect-first

A quote-first sender asks for an outcome; the quote works backwards to the amount to collect. A collect-first sender simply pays and the formulas of Sections 3 and 5 run forwards at conversion time.

```
target_out mode:  A_in = ceil_in( target_out / ( R_b × (1 - s) × (1 - r) ) )
fixed_in  mode:  client_net as in Section 3
```

ceil_in rounds *up* to the IN currency's minor unit so the sender never under-delivers the target. Example from the FRS: target 1,000 USD, 1 : 1 rate, 0.40 % fee all on the sender side → $A_{in} = 1004.016064 \rightarrow \mathbf{1004.02 \text{ USDT}}$; the same deal collect-first with 1,000 USDT yields 996.00 USD. Example at PT: to deliver a client net of 8,860.71 EUR, collect 9999.994583 → **9999.994583 USDT**.

Quote clocks. Freshness: the quote package must be marked sent within quote_freshness_min of the rate snapshot. Validity: quote_validity_min from sending. Grace: a further grace_min, honoured only if the market has drifted no more than grace_drift_pct: $|R_m(\text{now}) - R_m(\text{quote})| / R_m(\text{quote}) \leq \text{grace_drift_pct}$. Beyond grace, the late decision is a re-quote (new deal reference) or, where the project allows, honouring the old rate with Finance approval. The original quote figures never change.

7 · Variance thresholds

Check	Formula	When	Consequence
Partner rate breach	$\sigma = 1 - R_p / R_m$; breached if $\sigma > \text{avg_2w} \times (1 + \text{variance_threshold_rel})$, where avg_2w is the rolling two-week average of confirmed spreads for that partner × pair	Each new PARTNER_RATE_VERSION	Quoting and conversion on that partner × pair freeze until Finance approves with a reason, rejects, or a better rate is obtained (a better rate still outside the limit needs approval)
Minimum margin	$\text{margin_pct} \geq \text{min_margin_pct}$	At quote, and again at conversion	Below-margin quote proceeds only with Finance approval; at conversion a shortfall against the floor is logged
Conversion variance	$ \text{expected_out} - \text{actual_out} / \text{expected_out} \leq \text{rate_variation_rel}$ (THRESHOLD kind rate_variation_rel)	partner_calculates conversions	Exception opens; the deal does not wait
Grace drift	$ \Delta R_m / R_m \leq \text{grace_drift_pct}$	Payment arrives after validity, within grace	Grace applies only within drift; otherwise late decision
Basis variance on reroute	$A_{in} \times (R_{quote_basis} - R_{p,actual})$	Alternate partner chosen at conversion, fee on partner basis	Posted to VARIANCE on earnings; margin floor re-checked
Exposure	$\Sigma \text{ due} + \Sigma \text{ in-transit at a partner} \leq \text{exposure_ceiling}$	Each release, each day	Release blocked; under disburse_policy to_zero, balance after release must be < 1 unit

Breach examples with avg_2w = 0.50 % and variance_threshold_rel = 20 % (limit 0.600 %): PT at 0.8950 against market 0.9000 gives $\sigma = 0.556\%$ → not breached; PT at 0.8930 gives $\sigma = 0.778\%$ → **breached**.

8 · Referral commissions, rebates, markup decisions

```

fixed-percentage introducer:  s_total = s + ref_pct           (added to the sender share
before R_s)
                                referral_payable = round2( A_in × R_b × ref_pct )
share-of-earnings introducers: referral_payable_i = round2( earnings × share_i ), Σ
share_i ≤ 1
partner rebate:               rebate = round2( partner_cost × pct_of_partner_fee )
accrued per deal, reconciled monthly
return leg / balance conversion markup (FEE_DECISION):
    converted = round2( A_in × R_p )
    earnings = round2( converted × pct )    if decision = apply or override, else 0
    to client = converted - earnings

```

Example: on deal 3 (Section 9, V3) an introducer with a 25 % share of earnings is owed 11.20 EUR; Jeton's 32 % rebate on its 45.00 EUR fee is 14.40 EUR.

9 · Rounding and precision rules

Quantity	Rule
Rates	Stored and used at the pair's <code>rate_precision</code> : 5 decimals standard, 8 for sub-unit pairs such as USDT → BTC. Inversion for <code>in_per_out</code> pairs happens at full precision before storing.
Intermediate values	Exact decimals, no rounding.
Money amounts	Rounded half up to the currency's minor unit when stored: fiat 2 decimals, USDT 6, BTC 8.
Amount to collect (target-out)	Rounded <i>up</i> to the IN currency's minor unit.
Converted amounts and payouts	Truncated to whole units at conversion ($30.152 \rightarrow 30$). The dropped fraction is <code>rounding_residual</code> , stays client money in the balance, and is paid when accumulated dust reaches a whole unit. Disbursements pay whole units. Sub-unit differences never go to the shortfall ledger.
Cash collections	Rounded to the nearest <code>cash_rounding_unit</code> of the currency: 1000 for INR, 100 for SGD and USD, as configured per currency.
Percentages shown	Display only; the stored decimal is what the engine uses.
Comparisons	Thresholds are compared on exact decimals; "greater than" is strict.

10 · Reference vectors

Computed from the formulas above with round2 at 2 decimals for fiat and 8 for BTC. The engine must reproduce every cell. Vectors V1 and V3 are the deals used in the Ledger Posting Design.

Vector	Case												
V1	Collect-first, disclosed-rate partner (deal 1 in the ledger document)												
V2	Same deal priced on market basis												
V3	market + 0.50 % partner, market basis (deal 3 in the ledger document)												
V4	All fee on the sender side (FRS style), 1 : 1 rate												
V5	Receiver share only												
V6	Sub-unit pair, 8 dp rate (USDT → BTC)												

Vector	A_in	R_m	R_p	Basis	s / r	R_s	gross_out	fee_receiver	client_net	actual_out	partner_cost	earnings	margin
V1	10,000.00	0.9	0.895	partner	0.40 % / 0.60 %	0.89142	8,914.20	53.49	8,860.71	8,950.00	0.00	89.29	0.998 %
V2	10,000.00	0.9	0.895	market	0.40 % / 0.60 %	0.89640	8,964.00	53.78	8,910.22	8,950.00	0.00	39.78	0.444 %
V3	10,000.00	0.9	0.9	market	0.40 % / 0.60 % partner +0.50 %	0.89640	8,964.00	53.78	8,910.22	8,955.00	45.00	44.78	0.500 %
V4	1,004.02	1	1	market	0.40 % / 0.00 %	0.99600	1,000.00	0.00	1,000.00	1,004.02	0.00	4.02	0.400 %
V5	10,000.00	0.9	0.895	partner	0.00 % / 0.60 %	0.89500	8,950.00	53.70	8,896.30	8,950.00	0.00	53.70	0.600 %
V6	10,000.00	0.00001532	0.00001525	partner	0.40 % / 0.60 %	0.0000151890	0.15189000	0.00	0.15097866	0.15250000	0.00	0.00	0.998 %

Vector	Whole units paid	Rounding residual	fee_sender_part	gross_margin	Check
V1	8,860.00	0.71	35.80	89.29	gross_margin = partner_cost + earnings: ✓
V2	8,910.00	0.22	36.00	39.78	gross_margin = partner_cost + earnings: ✓
V3	8,910.00	0.22	36.00	89.78	gross_margin = partner_cost + earnings: ✓
V4	1,000.00	0.00	4.02	4.02	gross_margin = partner_cost + earnings: ✓
V5	8,896.00	0.30	0.00	53.70	gross_margin = partner_cost + earnings: ✓
V6	0.00	0.15097866	0.00	0.00	gross_margin = partner_cost + earnings: ✓

V6 is shown to 8 decimals; its whole-unit column is zero because a 0.15 BTC result has no whole unit, which is exactly the case the rounding rule must handle: the residual stays in balance until it can be paid.

11 · Acceptance

1. The engine reproduces every reference vector in Section 10 exactly, and every posting amount in the Ledger Posting Design.
2. For each project archetype, NPL's reference workbook (open question 11) is converted into vectors of the same shape and reproduced exactly; any difference is resolved by correcting either the workbook or this specification, never by tolerance.
3. The FRS worked figures (open question 9: 1,208 vs 1,225; 258,875 vs 300,000) are reconciled and added as golden cases.
4. Property checks on random inputs: debits equal credits per currency on every generated transaction; $\text{client_net} + \text{earnings} + \text{partner_cost} = \text{gross_converted}$; $\text{whole units} + \text{residual} = \text{client_net}$; entitlement conservation after any sequence of conversions, payouts, returns and balance conversions.

Draft v1.0 — for review with NPL. Figures are worked examples derived from the formulas, not NPL data, until the reference workbooks are supplied.